

SAFETY DATA SHEET

SECTION 1 IDENTIFICATION

1.1 PRODUCT IDENTIFIER	530 / HR-530
1.2 OTHER MEANS OF IDENTIFICATION	Potassium Iodide-Iodate Titrating Solution (1ml = 10 ppm as SO ₃ /50ml)
1.3 RECOMMENDED USE / RESTRICTIONS ON USE	Analytical Testing Reagent
1.4 SUPPLIERS DETAILS	WATER SCIENCES 3208 CAPRICE CT. FORT WAYNE, IN 46808 TEL: (260) 485-4655
1.5 EMERGENCY PHONE NUMBER	(800) 424-9300 CHEMTREC®

SECTION 2 HAZARD IDENTIFICATION

GENERAL HAZARD STATEMENT

To the best of our knowledge, this Safety Data Sheet is prepared to comply with US OSHA 29 CFR 1910.1200 and other governmental agencies, both domestic and foreign. Hazard and precautionary statements are based on maximum listed concentrations of each hazardous ingredient. Unlisted ingredients are not “hazardous” per the OSHA Hazard Communication Standard (29 CFR 1910.1200) and/or other governmental agencies

2.1 CLASSIFICATION OF THE SUBSTANCE OR MIXTURE

GHS CLASSIFICATION IN ACCORDANCE WITH (US) OSHA (29 CFR 1910.1200)

- Acute toxicity, oral, Category 4
- Skin corrosion / irritation, Category 4
- Eye damage / irritation, Category 2B

2.2 GHS LABEL ELEMENTS, INCLUDING PRECAUTIONARY STATEMENTS

PICTOGRAM



IRRITANT

SIGNAL WORD **WARNING**

HAZARD STATEMENTS

- H303 May be harmful if swallowed.
- H315 Causes mild skin irritation.
- H320 Causes eye irritation.

SECTION 2 HAZARD IDENTIFICATION – continued**PRECAUTIONARY STATEMENTS**

- P264 Wash hands thoroughly after handling.
- P270 Do not eat, drink or smoke when using this product.
- P280 Wear protective gloves/protective clothing/eye protection/face protection.

- P301+312 IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell.

- P301+330+331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
- P304+340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
- P305+351+338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do. Continue rinsing.

- P337+313 If eye irritation persists: Get medical advice/attention.
- P332+313 If skin irritation occurs: Get medical advice/attention.
- P362+364 Take off contaminated clothing and wash it before reuse.
- P501 Dispose of contents/container according to state and local regulations.

SECTION 3 COMPOSITION / INFORMATION ON INGREDIENTS**3.1 MIXTURES****Components****1. Potassium iodide**

Concentration	0-5% (weight)
Other Names/Synonyms	Kalii iodidum; Potassium iodide (KI)
EC no.	231-659-4
CAS no.	7681-11-0

Acute toxicity, oral, Category 4

Skin corrosion / irritation, Category 2

Eye damage / irritation, Category 2A

H302 Harmful if swallowed.

H315 Causes skin irritation.

H319 Causes serious eye irritation.

SECTION 3 COMPOSITION / INFORMATION ON INGREDIENTS – continued

2. Potassium iodate

Concentration	0-5% (weight)
Other Names/Synonyms	Iodic acid (HIO ₃), potassium salt (1:1); Iodic acid, potassium salt;
EC no.	231-831-9
CAS no.	7758-05-6

Oxidizing solids, Category 2

Skin corrosion / irritation, Category 2

Eye damage / irritation, Category 2A

Respiratory

H272 May intensify fire; oxidizer

H315 Causes skin irritation.

H319 Causes serious eye irritation.

H335 May cause respiratory irritation.

H336 May cause drowsiness or dizziness.

SECTION 4 FIRST AID MEASURES

4.1 DESCRIPTION OF FIRST AID MEASURES

General Advice	Consult a physician. Show this safety data sheet to the doctor in attendance.
If Inhaled	Remove person to fresh air and keep comfortable for breathing. Call a poison center or doctor if you feel unwell. Acute and delayed symptoms and effects: May cause respiratory irritation. Signs/symptoms may include irritation in the nose and throat, coughing.
In Case of Skin Contact	Take off immediately all contaminated clothing. Rinse skin with water/shower for at least 15 minutes. Call a poison center or doctor if irritation develops or persists. Wash contaminated clothing before reuse. Acute and delayed symptoms and effects: Causes mild skin irritation. Signs/symptoms may include localized redness, itching.
In Case of Eye Contact	Rinse cautiously with water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical attention/advice. Acute and delayed symptoms and effects: Causes eye irritation. Signs/symptoms may include redness, swelling, pain and tearing.
If Swallowed	Rinse mouth. Do NOT induce vomiting, call a poison center or doctor. If vomiting occurs naturally, have victim lean forward to reduce the risk of aspiration. Never give anything by mouth to an unconscious person. Acute and delayed symptoms and effects: May cause gastrointestinal irritation. Signs/symptoms may include abdominal pain, stomach upset, nausea, vomiting and diarrhea.

SECTION 4 FIRST AID MEASURES – Continued

4.2 MOST IMPORTANT SYMPTOMS/EFFECTS, ACUTE AND DELAYED

The most important known symptoms and effects are described in the labelling (See Section 2) and/or in Section 11.

4.3 INDICATION OF IMMEDIATE MEDICAL ATTENTION AND SPECIAL TREATMENT NEEDED, IF NECESSARY

After first aid, get appropriate in-plant, paramedic, or community medical expert.

SECTION 5 FIRE FIGHTING MEASURES

5.1 SUITABLE EXTINGUISHING MEDIA

Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

5.2 SPECIFIC HAZARDS ARISING FROM THE CHEMICAL

None known other than material can splatter above 100°C/212°F.

Potassium iodate: Hydrogen iodide, potassium oxides

5.3 SPECIAL PROTECTIVE ACTIONS FOR FIRE FIGHTERS

Avoid any skin contact. Effects of contact or inhalation may be delayed. Fire may produce irritating, corrosive and/or toxic gases. Runoff from fire control or dilution water may be corrosive and/or toxic and cause pollution. Wear positive pressure self-contained breathing apparatus (SCBA). Wear chemical protective clothing that is specifically recommended by the manufacturer. It may provide little or no thermal protection. Structural firefighters' protective clothing provides limited protection in fire situations ONLY; it is not effective in spill situations where direct contact with the substance is possible.

SECTION 6 ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Wear respiratory protection if necessary. Avoid breathing gas, mist, vapors, or spray. Ensure adequate ventilation. Evacuate personnel to safe areas. For personal protection see section 8.

6.2 Environmental precautions

Do not let product enter drains.

6.3 Methods and materials for containment and cleaning up

Stop leak if you can do it without risk. Sweep up and shovel into suitable containers for disposal.

Reference to other sections

For disposal see section 13.

SECTION 7 HANDLING AND STORAGE

7.1 Precautions for safe handling

Do not swallow. Wash thoroughly after handling. Do not eat, drink or smoke when using this product. See Section 8 for information on Personal Protective Equipment.

7.2 Conditions for safe storage, including any incompatibilities

Store away from incompatible materials. See Section 10 for information on Incompatible Materials. Keep out of the reach of children.

Specific end use(s)

Apart from the uses mentioned in section 1 no other specific uses are stipulated.

SECTION 8 EXPOSURE CONTROLS / PERSONAL PROTECTION

8.1 Control parameters

1. Potassium iodide (CAS: 7681-11-0 EC: 231-659-4)

TWA (Inhalation): 0.01 mg/m³ (ACGIH)

PEL: None Est. (OSHA)

USA. ACGIH Threshold Limit Values (TLV)/ Upper Respiratory Tract irritation, Hypothyroidism, Not classifiable as a human carcinogen

2. Potassium iodate (CAS: 7758-05-6 EC: 231-831-9)

PEL: None Est. (OSHA)

TLV®: None Est. (ACGIH)

8.2 Appropriate engineering controls

Use ventilation adequate to keep exposures (airborne levels of dust, fume, vapor, gas, etc.) below recommended exposure limits.

8.3 Individual protection measures, such as personal protective equipment (PPE)

PICTOGRAMS



Eye/face protection

Tightly fitting safety goggles. Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU). Ensure that eyewash stations and/or safety showers are close to the workstation location if working with concentrated product.

Skin protection

Wear protective gloves. Consult manufacturer specifications for further information.

Body protection

Wear protective clothing. Clothing with full length sleeves and pants should be worn. The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

SECTION 8 EXPOSURE CONTROLS / PERSONAL PROTECTION – continued

Respiratory protection

Where risk assessment shows air-purifying respirators are appropriate use a full-face respirator with multipurpose combination (US) or type ABEK (EN 14387) respirator cartridges as a backup to engineering controls. If the respirator is the sole means of protection, use a full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).

Thermal hazards

No data available.

Control banding approach

No data available.

Environmental exposure controls

Do not let product enter drains.

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

INFORMATION ON BASIC PHYSICAL AND CHEMICAL PROPERTIES

Appearance, form (physical state, color, etc.)	Clear color liquid
Odor	Slight odor
Odor Threshold	No data available
pH	8-10
Melting point / freezing point	Approx. 32°F
Initial boiling point and boiling range	212°F +/-
Flash point	No data available
Evaporation rate	1.00
Flammability (solid, gas)	Non-flammable
Upper/lower flammability limits	No data available
Upper/lower explosive limits	No data available
Vapor pressure	No data available
Vapor density	No data available
Relative density	No data available
Solubility	Completely soluble in water
Partition coefficient: n-octanol/water	No data available
Auto-ignition temperature	No data available
Decomposition temperature	No data available
Viscosity	No data available
Explosive properties	No data available
Oxidizing properties	No data available

SECTION 10 STABILITY AND REACTIVITY

10.1 STABILITY AND REACTIVITY

Contact with incompatible materials. Sources of ignition. Exposure to heat.

10.2 CHEMICAL STABILITY

Stable under normal storage conditions.

10.3 POSSIBILITY OF HAZARDOUS REACTIONS

No data available.

10.4 CONDITIONS TO AVOID

Excessive temperature contact with any type of reactive chemicals.

10.5 INCOMPATIBLE MATERIALS

Avoid contact with Strong Alkalis, Oxidizers or any other type of reactive material.
Potassium iodide: Strong reducing agents, Nickel, Strong acids, and its alloys, Steel (all types and surface treatments), Aluminum, Alkali metals, Brass, Magnesium, Zinc, cadmium, Copper Potassium iodate: Strong reducing agents, Powdered metals, Incompatibility: mixtures of iodates with finely divided aluminum, arsenic, copper, carbon, phosphorous (red or white) sulfur; hydrides of alkali and alkaline earth metals; sulfides of antimony, arsenic, copper or tin, metal cyanides, thiocyanates or impure manganese dioxide may react violently or explosively, either spontaneously (especially in the presence of moisture) or on initiation by heat, friction impact, sparks, or addition of sulfuric acid

10.6 HAZARDOUS DECOMPOSITION PRODUCTS

There are no known hazardous decomposition products for this material unless the material is burned, in which case undetermined evolution of toxic gases may occur.

Potassium iodate: Other decomposition products - no data available.

In the event of fire: see section 5

SECTION 11 TOXICOLOGICAL INFORMATION

Information on toxicological effects

Acute toxicity

Like Routes of Exposure: Eye contact, Skin contact, Inhalation, Ingestion

Components

Potassium iodate: LDLO Oral - Mouse - 531 mg/kg

Potassium iodate: LDLO Oral - Guinea pig - 400 mg/kg

Potassium iodide: LD50 Oral - Mouse - 1,000 mg/kg

Symptoms (including delayed and immediate effects):

Inhalation: May cause respiratory irritation. Signs/symptoms may include burning pain in the nose and throat, coughing

Ingestion: May cause gastrointestinal irritation. Signs/symptoms may include abdominal pain, stomach upset, nausea, vomiting and diarrhea.

SECTION 11 TOXICOLOGICAL INFORMATION – continued**Skin corrosion / irritation**

Causes skin irritation. Signs/symptoms may include localized redness, swelling, and itching.

Potassium iodate: LD50 Intraperitoneal - Mouse - 136 mg/kg

Serious eye damage / irritation

Causes eye irritation. Signs/symptoms may include cloudy appearance of the cornea, severe pain, tearing.

Respiratory or skin sensitization

No data available on product

Components: No data available.

Germ cell mutagenicity

No data available.

Carcinogenicity

This product is or contains a component that is not classifiable as to its carcinogenicity based on its IARC, ACGIH, NTP, or EPA classification

OSHA: No component of this product present at levels greater than or equal to 0.1% is identified as a carcinogen or potential carcinogen by OSHA.

Reproductive toxicity

No data available.

Summary of evaluation of the CMR properties

No data available.

STOT-single exposure

No data available.

STOT-repeated exposure

No data available.

Aspiration hazard

No data available.

Additional information: No data available.

SECTION 12 ECOLOGICAL INFORMATION**Toxicity**

No data available.

Potassium iodide: LC50 - Oncorhynchus mykiss (rainbow trout) - 2,190 mg/l - 96 h

Potassium iodide: EC50 - Daphnia magna (water flea) - 2.7 mg/l - 24 h

Persistence and degradability

No data available on product

Bioaccumulative potential

No data available on product

SECTION 12 ECOLOGICAL INFORMATION – continued

Mobility in soil

No data available.

Results of PBT and vPvB assessment

PBT/vPvB assessment not available as chemical safety assessment not required/not conducted

Other adverse effects

No data available.

SECTION 13 DISPOSAL CONSIDERATIONS

Disposal of the product

Disposal should be in accordance with applicable Federal, State and local laws and regulations. Local regulations may be more stringent than State or Federal requirements.

Disposal of contaminated packaging

Dispose of as unused product.

Waste treatment

Contact your supplier or a licensed contractor for detailed recommendations. Follow applicable federal, state and local regulations.

Sewage disposal

The product should not be allowed to enter drains, water courses or the soil.

Other disposal recommendations

Triple rinse the empty containers with water before disposal to reconditioned or land fill or garbage.

SECTION 14 TRANSPORT INFORMATION

DOT (US)

UN Number: NA

Class: NA

Packing Group: NA

Proper Shipping Name: NON-REGULATED MATERIALS

Reportable quantity (RQ): NA

Marine pollutant: No

Poison inhalation hazard: No

IMDG

UN Number:

Class:

Packing Group:

EMS Number:

Proper Shipping Name:

SECTION 14 TRANSPORT INFORMATION – continued

IATA

UN Number:

Class:

Packing Group:

Proper Shipping Name:

SECTION 15 REGULATORY INFORMATION

15.1 Safety, health and environmental regulations specifically for the product in question

SARA 302 Components

SARA 302: No chemicals in this material are subject to the reporting requirements of SARA Title III, Section 302.

SARA 311/312 Hazards

Chronic Health Hazard

SARA 313 Components

SARA 313: This material does not contain any chemical components with known CAS numbers that exceed the threshold (De Minimis) reporting levels established by SARA Title III, Section 313.

Toxic Substances Control Act (TSCA) Inventory

All components of this product are in compliance with the inventory listing requirements of the U.S. Toxic Substances Control Act (TSCA) Chemical Substance Inventory

Comprehensive Environmental Response, Compensation, and Liability Act of 1980 (CERCLA):

This material, as supplied, does not contain one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) RQ: Not listed

California Prop. 65 Components

This product does not contain any chemicals known to State of California to cause cancer, birth defects, or any other reproductive harm.

Massachusetts Right To Know Components

No components are subject to the Massachusetts Right to Know Act.

New Jersey Right To Know Components

Potassium iodide: CAS number: 7681-11-0

Potassium iodate: CAS number: 7758-05-6

Pennsylvania Right To Know Components

Potassium iodide: CAS number: 7681-11-0

Potassium iodate: CAS number: 7758-05-6

HMIS RATING

HEALTH	1
FLAMMABILITY	0
PHYSICAL HAZARD	0
PERSONAL PROTECTION	B

NFPA RATING

HEALTH	1
FLAMMABILITY	0
INSTABILITY HAZARD	0
SPECIAL HAZARD	W

SECTION 16 OTHER INFORMATION

16.1 Further information/disclaimer

DISCLAIMER: The information above is believed to be accurate and represents the best information currently available to us. However, we make no warranty of merchantability or any other warranty, express or implied, with respect to such information, and we assume no liability resulting from its use. Users should make their own investigation to determine the suitability of information for their particular purposes. In no event shall manufacturer be liable for any claims, losses, or damages of any third party or for lost profits or any special, indirect, incidental, consequential or exemplary damages, whatsoever arising, even if manufacturer has been advised of the possibility of such damages.

Date of Preparation/Revision: 6/2024